

Experimental Setup

Does the explicit homogeneous projector router realize the derived long-wavelength cone, including finite-band and degeneracy controls?

Experiment: gross.router-propagation.v1 Model: signal-space.ss-ops-1.router-linear.v1
Runtime configuration: fixtures/research/gross-test-03.json
Plan lock: 028c20b230d8b4ffed30d7aaa33ea7c769c18b20387ff807557e22cc0fb09d47

MODEL AND NUMERICAL SETUP

Action: Exact zero-wave linearization of SS OPS 1 at $\kappa=\pi/2$, $\lambda=0$: $S_i \psi(m)=P_i+ \psi(m-e_i)+P_i- \psi(m+e_i)$, $U=S_3 S_2 S_1$. Memories homogeneous and stationary; this sector registration is not a new full dynamics.
Units: $\ell=\tau_c=1$; $q=k \ell$, $\text{phase}=\omega \tau_c$, group velocity in ℓ/τ_c . Neither conversion scale is predicted.
Domain: Three fixed unit triads, both routing orders, all 26 normalized nonzero vectors in $\{-1,0,1\}^3$, radii $0.02/0.04/0.08/0.16$. Full periodic zone $[-\pi,\pi]^3$ on 16 and 32 points per axis, including explicit corner and half- π probes.
Boundaries: Infinite homogeneous lattice via exact Fourier blocks; random periodic boxes 8^3 and 12^3 verify stencil and norm, not a continuum PDE extrapolation. No outgoing boundary or time integrator error.
Initial Data: Triads fixed in configuration. Periodic random complex fields normalized globally using runtime PCG64 sampling stream; rotate all projectors about $(1,2,3)/\sqrt{14}$ by 0.731 rad.
Solver: Compose displacement coefficients from the three explicit conditional shifts. Independently compare scipy matrix exponentials, analytic quaternion product and full real-space shifts/FFT. Principal positive quasiphase in $[0,\pi]$; both eigenbands plus/minus phase, branch derivatives undefined at touchings.
Tolerances: Matrices and norm $<1e-10$. Fine cone residual <0.02 and adjacent halving ratio <0.65 above roundoff floor. Derivative steps $1e-5, 5e-6$, reference and refinement $<1e-6$. Full details in each criterion.

COMPETING PREDICTIONS

derived-cone: Independent triads give nondegenerate Gram cones; collinear gives rank one. Order and directional asymmetry are finite-band terms, suppressed at small wave number.
incorrect-routing: Stencil/exponential disagreement, missing rank loss or residuals that fail to shrink falsify the implementation or the proposed leading cone within this benchmark.

CONTROLS

independent-reference: Expose routing signs, order errors and normalization errors; method: Exponential product, independent quaternion algebra, periodic random field FFT and common projector rotation.
degeneracy-order: Separate internal algebra from supplied propagation geometry; method: Collinear triad, reversed gate order and opposite wavevectors; preserve exact nodal cases.
resolution: Discriminate finite-band corrections and numerical errors; method: Four prescribed radii, two derivative steps, nested full-zone grids and two periodic boxes. Do not fit coefficients to the same measured data.

PREREGISTERED DECISIONS

coverage (Save all 3 triads, 2 orders, 26 signed directions, 4 radii, both zone and box sizes, rotation and derivative comparisons.)
Pass: Save all 3 triads, 2 orders, 26 signed directions, 4 radii, both zone and box sizes, rotation and derivative comparisons.
Fail: Complete finite evidence violates the declared gate.

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Unresolved: Missing, nonfinite or incomplete evidence; not a pass.
Uncertainty: Independent exact formulas, radius/derivative refinement and nested zone grids;
deterministic tolerances, not statistical confidence.

fourier-circuit (Stencil matrix agrees with independent matrix exponentials and real-space fields with FFT reference to $<1e-10$.)
Pass: Stencil matrix agrees with independent matrix exponentials and real-space fields with FFT reference to $<1e-10$.
Fail: Complete finite evidence violates the declared gate.
Unresolved: Missing, nonfinite or incomplete evidence; not a pass.
Uncertainty: Independent exact formulas, radius/derivative refinement and nested zone grids;
deterministic tolerances, not statistical confidence.

conservation (Matrix unitarity, $\det U=1$ and total wave norm errors $<1e-10$.)
Pass: Matrix unitarity, $\det U=1$ and total wave norm errors $<1e-10$.
Fail: Complete finite evidence violates the declared gate.
Unresolved: Missing, nonfinite or incomplete evidence; not a pass.
Uncertainty: Independent exact formulas, radius/derivative refinement and nested zone grids;
deterministic tolerances, not statistical confidence.

continuum (Maximum $|\text{phase}^2 - q \cdot G \cdot q|/|q|^2 < 0.02$ at radius 0.02; each adjacent fine/coarse error envelope ratio < 0.65 unless coarse error $\leq 1e-10$.)
Pass: Maximum $|\text{phase}^2 - q \cdot G \cdot q|/|q|^2 < 0.02$ at radius 0.02; each adjacent fine/coarse error envelope ratio < 0.65 unless coarse error $\leq 1e-10$.
Fail: Complete finite evidence violates the declared gate.
Unresolved: Missing, nonfinite or incomplete evidence; not a pass.
Uncertainty: Independent exact formulas, radius/derivative refinement and nested zone grids;
deterministic tolerances, not statistical confidence.

triad-rank (Gram matrix ranks [3,3,1] for orthogonal, oblique, collinear, with rank tolerance $1e-10$.)
Pass: Gram matrix ranks [3,3,1] for orthogonal, oblique, collinear, with rank tolerance $1e-10$.
Fail: Complete finite evidence violates the declared gate.
Unresolved: Missing, nonfinite or incomplete evidence; not a pass.
Uncertainty: Independent exact formulas, radius/derivative refinement and nested zone grids;
deterministic tolerances, not statistical confidence.

order-control (Orthogonal trace matches the signed triple-sine formula to $<1e-10$; maximum order phase difference $>1e-4$ for each independent triad.)
Pass: Orthogonal trace matches the signed triple-sine formula to $<1e-10$; maximum order phase difference $>1e-4$ for each independent triad.
Fail: Complete finite evidence violates the declared gate.
Unresolved: Missing, nonfinite or incomplete evidence; not a pass.
Uncertainty: Independent exact formulas, radius/derivative refinement and nested zone grids;
deterministic tolerances, not statistical confidence.

rotation (Common SU(2) conjugation agrees with rotated-projector stencil to $<1e-10$.)
Pass: Common SU(2) conjugation agrees with rotated-projector stencil to $<1e-10$.
Fail: Complete finite evidence violates the declared gate.
Unresolved: Missing, nonfinite or incomplete evidence; not a pass.
Uncertainty: Independent exact formulas, radius/derivative refinement and nested zone grids;

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deterministic tolerances, not statistical confidence.

group-refinement (Central steps $1e-5$ and $5e-6$ agree with independent analytic quaternion derivative and each other to $<1e-6$ away from band touchings (vector norm $>1e-6$).)

Pass: Central steps $1e-5$ and $5e-6$ agree with independent analytic quaternion derivative and each other to $<1e-6$ away from band touchings (vector norm $>1e-6$).

Fail: Complete finite evidence violates the declared gate.

Unresolved: Missing, nonfinite or incomplete evidence; not a pass.

Uncertainty: Independent exact formulas, radius/derivative refinement and nested zone grids; deterministic tolerances, not statistical confidence.

full-zone (Both $16^3/32^3$ periodic-zone samples agree with quaternion reference to $<1e-10$; nested samples agree $<1e-10$; explicit node probes agree $<1e-10$; orthogonal grids detect >1 zero node and ≥ 1 pi node.)

Pass: Both $16^3/32^3$ periodic-zone samples agree with quaternion reference to $<1e-10$; nested samples agree $<1e-10$; explicit node probes agree $<1e-10$; orthogonal grids detect >1 zero node and ≥ 1 pi node.

Fail: Complete finite evidence violates the declared gate.

Unresolved: Missing, nonfinite or incomplete evidence; not a pass.

Uncertainty: Independent exact formulas, radius/derivative refinement and nested zone grids; deterministic tolerances, not statistical confidence.

VISUAL QUESTIONS

cone-sections [evidentiary]: Which projector arrangements yield an isotropic, anisotropic or degenerate cone?

View: Three exact low-frequency contours over leading Gram contours in the $qz=0$ plane.; observables: phase, Gram matrix, residuals

derived-cone: Independent triads give nondegenerate Gram cones; collinear gives rank one. Order and directional asymmetry are finite-band terms, suppressed at small wave number.

incorrect-routing: Stencil/exponential disagreement, missing rank loss or residuals that fail to shrink falsify the implementation or the proposed leading cone within this benchmark.

Controls: independent-reference, degeneracy-order, resolution; uncertainty: Exact saved samples with explicit phase-branch and finite-grid limits; no fitted metric or smoothing.

Data: reports/<report-id>/plot-data/surfaces.npz, reports/<report-id>/plot-data/geometry.json;

transforms: See saved figure specification; low-q contours interpolate a 101×101 grid, full-zone sections use saved grid samples; exact data retained.

continuum-convergence [evidentiary]: Does the actual circuit approach its predicted cone as wavelength grows?

View: Maximum squared-phase residual by radius and triad, for both orders.; observables: phase, Gram matrix, residuals

derived-cone: Independent triads give nondegenerate Gram cones; collinear gives rank one. Order and directional asymmetry are finite-band terms, suppressed at small wave number.

incorrect-routing: Stencil/exponential disagreement, missing rank loss or residuals that fail to shrink falsify the implementation or the proposed leading cone within this benchmark.

Controls: independent-reference, degeneracy-order, resolution; uncertainty: Exact saved samples with explicit phase-branch and finite-grid limits; no fitted metric or smoothing.

Data: reports/<report-id>/plot-data/convergence.csv; transforms: See saved figure specification; low-q contours interpolate a 101×101 grid, full-zone sections use saved grid samples; exact data retained.

order-and-direction [evidentiary]: Is opposite-direction asymmetry a finite-band order effect or leading drift?

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View: Signed order differences and omega odd divided by $|q|$ for a fixed body diagonal.; observables: phase, Gram matrix, residuals
derived-cone: Independent triads give nondegenerate Gram cones; collinear gives rank one. Order and directional asymmetry are finite-band terms, suppressed at small wave number.
incorrect-routing: Stencil/exponential disagreement, missing rank loss or residuals that fail to shrink falsify the implementation or the proposed leading cone within this benchmark.
Controls: independent-reference, degeneracy-order, resolution; uncertainty: Exact saved samples with explicit phase-branch and finite-grid limits; no fitted metric or smoothing.
Data: reports/<report-id>/plot-data/dispersion.csv; transforms: See saved figure specification; low-q contours interpolate a 101×101 grid, full-zone sections use saved grid samples; exact data retained.

full-zone-spectrum [evidentiary]: What additional low and pi-quasifrequency modes are hidden outside the origin?

View: Six full-zone sections at $q_z=0$ and $-\pi/2$ with sampled node counts at both resolutions.;
observables: phase, Gram matrix, residuals
derived-cone: Independent triads give nondegenerate Gram cones; collinear gives rank one. Order and directional asymmetry are finite-band terms, suppressed at small wave number.
incorrect-routing: Stencil/exponential disagreement, missing rank loss or residuals that fail to shrink falsify the implementation or the proposed leading cone within this benchmark.
Controls: independent-reference, degeneracy-order, resolution; uncertainty: Exact saved samples with explicit phase-branch and finite-grid limits; no fitted metric or smoothing.
Data: reports/<report-id>/plot-data/surfaces.npz, reports/<report-id>/plot-data/zones.csv; transforms: See saved figure specification; low-q contours interpolate a 101×101 grid, full-zone sections use saved grid samples; exact data retained.

group-directions [evidentiary]: Does the energy-transport direction follow the wavevector in the oblique case?

View: Measured and leading-cone group vectors for xy directions at smallest radius.; observables: phase, Gram matrix, residuals
derived-cone: Independent triads give nondegenerate Gram cones; collinear gives rank one. Order and directional asymmetry are finite-band terms, suppressed at small wave number.
incorrect-routing: Stencil/exponential disagreement, missing rank loss or residuals that fail to shrink falsify the implementation or the proposed leading cone within this benchmark.
Controls: independent-reference, degeneracy-order, resolution; uncertainty: Exact saved samples with explicit phase-branch and finite-grid limits; no fitted metric or smoothing.
Data: reports/<report-id>/plot-data/groups.csv; transforms: See saved figure specification; low-q contours interpolate a 101×101 grid, full-zone sections use saved grid samples; exact data retained.

numerical-audit [evidentiary]: Do explicit port shifts conserve amplitude and agree with the independent Fourier reference?

View: Norm/FFT residuals for both finite periodic boxes and global matrix/refinement checks.;
observables: phase, Gram matrix, residuals
derived-cone: Independent triads give nondegenerate Gram cones; collinear gives rank one. Order and directional asymmetry are finite-band terms, suppressed at small wave number.
incorrect-routing: Stencil/exponential disagreement, missing rank loss or residuals that fail to shrink falsify the implementation or the proposed leading cone within this benchmark.
Controls: independent-reference, degeneracy-order, resolution; uncertainty: Exact saved samples with explicit phase-branch and finite-grid limits; no fitted metric or smoothing.
Data: reports/<report-id>/plot-data/fields.csv, reports/<report-id>/plot-data/summary.json; transforms: See saved figure specification; low-q contours interpolate a 101×101 grid, full-zone sections use saved

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grid samples; exact data retained.

RESOURCE LIMITS

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{"max_memory_mb": 512, "max_output_mb": 32, "max_wall_seconds": 120}
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PROHIBITED INFERENCES

No all-sector shared geometry: physical memory modes remain stationary at this vacuum and are not included in this wave-sector spectrum.

No autonomous clock, detector record, fermion species, protected topology or emergent spacetime inference.

Finite-band directional asymmetry is not a leading identity drift; other backgrounds and laws remain open under Tests 4 and 11.

Full-zone grid inspection detects extra nodes but is not a completeness or topological-charge proof.